SIMATS ENGINEERING

CO2- PROBLEM SOLVING ASSIGNMENT

Course: ECA1803– Optoelectronics and Optical Communication
Assessment: CO2 Problem-Solving Assignment

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STANDARDS-COMPLIANT LASER TRANSMITTER DESIGN, WAVELENGTH CONTROL AND NONLINEARITY MANAGEMENT FOR AN 8-CHANNEL DWDM LINK

1. PROBLEM REQUIREMENT AND SYSTEM ANALYSIS

1.1 Problem Statement

The proposed transmitter must provide sufficient optical power, high modulation bandwidth, narrow spectral linewidth, stable wavelength operation and reliable transmission while minimizing nonlinear impairments. It must also consider optical safety, applicable interface standards, dispersion compensation and photonic integration.

The objective of this assignment is to design and evaluate an appropriate optical transmitter for an eight-channel DWDM link operating around 1550 nm with 100 GHz channel spacing. Dense Wavelength Division Multiplexing (DWDM) is an optical communication technique that transmits multiple independent data channels simultaneously through a single optical fibre using different optical wavelengths.

Each channel carries 25 Gb/s Non-Return-to-Zero (NRZ) data through six amplified spans, each 80 km long, of standard single-mode fibre. The total transmission distance is therefore:

Total distance = 6 × 80 = **480 km**

The principal design challenge is to select an optical source and transmitter configuration that satisfy these requirements while maintaining acceptable signal quality across the complete link.

1.2 System Requirements

Parameter	**Specified value**
Number of DWDM channels	8
Operating wavelength	Approximately 1550 nm

Channel spacing	100 GHz, approximately 0.8 nm
Data rate	25 Gb/s per channel
Modulation format	NRZ
Fibre length	6 × 80 km = 480 km
Fibre attenuation	0.2 dB/km
Nonlinear coefficient	1.3 $W^{-1}km^{-1}$
Effective fibre area	80 μm^2
Launch power	2 mW per channel
Laser threshold current	12 mA
Bias current	40 mA
Slope efficiency	0.3 mW/mA
Modulation bandwidth	20 GHz
Ambient temperature variation	40 °C
Required wavelength tolerance	±0.1 nm

1.3 Problem Requirement Statement

It must maintain the assigned wavelengths within the required tolerance, provide adequate optical power and modulation bandwidth, and control dispersion, nonlinear effects and crosstalk. The design must incorporate appropriate wavelength stabilization, optical safety measures, applicable standards verification and a practical photonic integration strategy. The proposed transmitter must generate eight wavelength-separated optical channels with 25 Gb/s NRZ data per channel and approximately 200 Gb/s aggregate data capacity.

2. OPTICAL SOURCE SELECTION

2.1 Emission Wavelength and Quantum Efficiency

Emission wavelength is the wavelength of light generated by the laser diode. In DWDM systems, each channel operates at a different wavelength to transmit multiple signals through a single optical fibre.

For this project, the C-band wavelength region (1530–1565 nm) is selected because it offers low fibre attenuation and supports DWDM transmission.

The frequency and wavelength relationship is:

$$\lambda = \frac{c}{f}$$

2.2 Comparison of Optical Sources

Optical source	**Characteristics**	**Suitability for this link**
LED	Broad spectrum, low optical power and limited high-speed performance	Unsuitable for long-distance DWDM
Fabry–Perot laser	Multiple longitudinal modes and comparatively broad spectrum	Less suitable because of spectral overlap and dispersion concerns
DBR laser	Distributed Bragg reflector provides wavelength-selective feedback	Suitable for narrow-linewidth, wavelength-controlled applications
DFB laser	Distributed feedback supports stable, single-longitudinal-mode operation	Highly suitable for high-speed DWDM
VCSEL	Compact, efficient and suitable for many short-reach applications	Possible alternative for suitable wavelength and fibre-interface designs
Quantum-dot laser	Potential advantages in temperature stability and threshold behaviour, depending on device design	Promising alternative if the required wavelength and performance are demonstrated

A narrow linewidth and high side-mode suppression ratio (SMSR) reduce spectral spreading and unwanted channel interference. The source must support a stable single-mode spectrum, sufficient modulation speed and appropriate fibre coupling. However, actual linewidth and SMSR values must be obtained from the selected device datasheet.

2.3 Selected Optical Source

Design Decision 1: I recommend a temperature-controlled DFB laser source for each DWDM wavelength channel.

The selection is justified by the following considerations:

1. Its single-mode operation is suitable for wavelength-separated DWDM transmission.
2. It supports high-speed modulation when the selected device has adequate bandwidth.
3. Its narrow spectral linewidth can reduce phase-noise and stimulated Brillouin scattering concerns.
4. Its wavelength can be stabilized using temperature control and appropriate drive-current regulation.
5. It is suitable for integration into multi-channel optical transmitter assemblies.

A DBR laser is a technically credible alternative when wider wavelength tuning or particular wavelength-selection requirements make it preferable. The final device must be verified against the required power, linewidth, SMSR and temperature specifications.

3. MODULATION, OUTPUT POWER AND WAVELENGTH CONTROL

3.1 Optical Output Power

Optical output power is the amount of optical energy emitted by a laser diode per unit time. It is an important parameter in an 8-channel DWDM communication system because it determines the strength of the optical signal transmitted through the fibre.

In the proposed system, Distributed Feedback (DFB) laser diodes are used as optical sources to generate stable light signals at different wavelengths. Each laser diode provides the required optical power for its respective DWDM channel.

This is the calculated laser output power using the supplied linear slope-efficiency model. It is not automatically the power launched into the fibre because coupling losses, modulation losses, connectors and attenuation may reduce the available power.

The specified launch power is 2 mW per channel, equivalent to:

$P_{launch}(dBm) = 10 \log_{10}(2) =$ **3.01 dBm per channel**

Thus, the laser output estimate exceeds the specified launch level, leaving a nominal power difference of 6.23 dB before accounting for actual component losses. An optical attenuator or suitable power-control stage may be used where necessary.

3.2 Modulation Bandwidth

The approximate bandwidth requirement for NRZ data is:

$B \approx 0.7R_b$

Given $R_b = 25$ Gb/s:

$B \approx 0.7 \times 25 = 17.5$ GHz

The specified modulation bandwidth is 20 GHz.

Bandwidth margin = 20 − 17.5 = 2.5 GHz

Percentage margin = (2.5 / 17.5) × 100 = 14.29%

The candidate bandwidth is adequate according to the supplied approximation, with a 2.5 GHz margin. Nevertheless, practical performance must also be verified through the complete electro-optical response, rise time, chirp and eye-diagram measurements.

3.3 Carrier and Photon Rate Equations

Carrier and photon rate equations describe the operation of a semiconductor laser diode. They explain how electrical energy supplied to the laser is converted into optical energy in the form of photons.

In the proposed 8-channel DWDM system, DFB laser diodes are used to generate optical signals at different wavelengths. These equations help analyse carrier injection, photon generation, optical output power and laser modulation performance.

Here, N represents carrier density, S represents photon density, I is injection current, q is electronic charge, V is active-region volume, R(N) is the carrier recombination rate, Γ is the optical confinement factor, v_g is group velocity, g is optical gain, β_sp is the spontaneous-emission coupling factor and τ_p is photon lifetime.

$$\boxed{\frac{dN}{dt} = \frac{I}{qV} - \frac{N}{\tau_n} - \Gamma v_g g(N) S}$$

3.4 Direct Versus External Modulation

It offers better control of chirp and is appropriate when transmission distance and spectral stability are important.

External modulation uses a separate device, such as a Mach–Zehnder modulator, to encode the data while the laser operates at a more stable bias point. Direct modulation changes the laser injection current to encode data. It is comparatively simple and economical but can produce frequency chirp and interact with chromatic dispersion.\

3.5 Temperature Stabilization

Injection-current control can provide fine wavelength tuning, but it also changes carrier density and may affect output power and chirp. TEC control is recommended as the primary stabilization mechanism, with current control used for bias regulation and carefully calibrated fine tuning.

The uncontrolled wavelength drift is:

Δλ = wavelength drift per °C × temperature variation

Δλ = 0.1 × 40 = **4 nm**

The drift is five times the approximate 0.8 nm channel spacing. It is also 40 times the permitted absolute wavelength deviation of 0.1 nm.

The maximum temperature deviation corresponding to ±0.1 nm is:

ΔT = 0.1 nm / (0.1 nm/°C) = **±1 °C**

A thermoelectric cooler (TEC) with a temperature sensor and feedback controller should therefore stabilize the laser temperature to approximately ±1 °C under the given linear drift model. Additional margin may be needed for sensor error, package gradients and other

Design Decision 2: I recommend external modulation with TEC-based wavelength stabilization because it offers better control of chirp and wavelength drift for the 480 km DWDM link.

4. NONLINEAR IMPAIRMENTS AND THEIR CONTROL

4.1 Effective Fibre Length

Effective fibre length is the actual effective distance over which optical power propagates inside an optical fibre while considering the loss caused by fibre attenuation.

In an 8-channel DWDM communication system, effective fibre length is an important parameter because optical power decreases as the signal travels through the fibre. It is also used to analyse nonlinear effects such as Self-Phase Modulation (SPM), Cross-Phase Modulation (XPM) and Four-Wave Mixing (FWM).

$$\boxed{L_{eff} = \frac{1 - e^{-\alpha L}}{\alpha}}$$

4.2 Self-Phase Modulation

Self-Phase Modulation (SPM) is a nonlinear optical effect that occurs when a high-power optical signal travels through an optical fibre. It is caused by a change in the refractive index of the fibre due to the intensity of the optical signal.

In the proposed 8-channel DWDM system, SPM affects the phase of each transmitted optical channel. This can cause spectral broadening, signal distortion and reduced communication performance.

1. Working Principle

1. The DFB laser diode generates an optical signal.
2. The optical signal travels through the single-mode fibre.
3. High optical intensity changes the refractive index of the fibre.
4. This change produces a phase shift in the transmitted signal.
5. The phase shift causes spectral broadening, which may affect signal quality.

4.3 Cross-Phase Modulation and Four-Wave Mixing

Its impact depends on channel power, pulse overlap, walk-off and fibre dispersion. Four-wave mixing (FWM) involves interactions among optical frequencies that generate additional frequency components. Cross-phase modulation (XPM) occurs when the intensity of one channel changes the phase of another channel through the Kerr effect. The actual severity of XPM and FWM cannot be quantified from channel spacing alone. To control these

impairments, the design should use controlled launch power, suitable channel spacing, dispersion-aware system planning and appropriate optical filtering.

These components can interfere with useful DWDM channels. The 100 GHz channel spacing establishes the frequency relationships, while phase matching and dispersion influence FWM efficiency.

4.4 Eye Safety and Protective Earthing

Practical safety measures include enclosed optical paths, connector interlocks where appropriate, laser warning labels, power monitoring and safe fibre inspection procedures. Optical connectors must never be viewed directly when a laser may be active. The assessment must consider accessible emission, wavelength, beam characteristics, exposure duration, enclosure design and the applicable standard. The total launched optical power is 16 mW, equivalent to approximately 12.04 dBm. This value alone does not establish an ANSI Z136 laser safety classification. The actual accessible emission and laser classification must be determined before deployment. For preliminary screening only, if a project adopts an illustrative 10 mW accessible-output limit, 16 mW would exceed that assumed limit. This is not a universal ANSI Z136 limit and must not be represented as proof of compliance or non-compliance.

Protective earthing: The transmitter chassis, metal enclosure and exposed conductive parts of mains-powered equipment must be connected to protective earth according to the equipment manufacturer's instructions and applicable electrical codes. Protective earth must not be replaced by signal ground, an optical-fibre shield or an improvised connection. Fibre itself does not provide a protective-earth path. Earth-electrode dimensions and earth resistance cannot be specified from this assignment because the installation supply, soil conditions and applicable electrical requirements are not provided.

Design Decision 3: I recommend a controlled-launch-power strategy, dispersion-aware channel planning, suitable filtering and formal optical safety verification. The calculated SBS threshold indicates that launch power and linewidth require particular attention. Any mitigation must preserve adequate received signal quality.

5. SOLITON FEASIBILITY, DIGITAL BACKPROPAGATION AND INTEGRATION

5.1 Dispersion Length

Soliton transmission requires a balance between anomalous chromatic dispersion and Kerr nonlinearity.

The dispersion length is:

For a fundamental soliton, $L_{NL} = L_D$.

Since $L_{NL} = 1/(\gamma P_0)$:

$P_0 = 1/(\gamma L_D)$

$P_0 = 1/(1.3 \times 11.25)$

$P_0 = 0.06838$ W = **68.38 mW**

5.2 Fundamental Soliton Peak Power

This is the calculated peak power required for a fundamental soliton under the supplied model and assumed anomalous-dispersion conditions.

$L_D = T_0^2 / |\beta_2|$

Given $T_0 = 15$ ps and $|\beta_2| = 20$ ps²/km:

$L_D = 15^2 / 20 = 225 / 20$

L_D = 11.25 km

5.3 Soliton Order at the Specified Launch Power

At $P_0 = 2$ mW:

$L_{NL} = 1/(\gamma P_0)$

$L_{NL} = 1/(1.3 \times 0.002)$

L_{NL} = 384.62 km

The soliton order is:

$N^2 = L_D/L_NL$

$N = \sqrt{(11.25/384.62)}$

$N \approx 0.171$

Since N is substantially below unity, the specified launch power does not satisfy the fundamental-soliton condition. Therefore, soliton transmission is **not recommended under the given operating conditions**. The calculated peak-power requirement also assumes that the stated pulse width is the soliton parameter T_0 and that the required dispersion regime is available.

5.4 Digital Backpropagation

Its limitations include computational complexity, sensitivity to model mismatch and the need for suitable sampled optical-field information. Full multi-channel compensation is especially demanding because interactions among all eight channels must be represented. Digital backpropagation (DBP) numerically reverses propagation effects by modelling chromatic dispersion and Kerr nonlinearity. It can improve signal quality when accurate channel information and suitable receiver processing are available.

For the stated 25 Gb/s NRZ link, full-link DBP should not be assumed to be directly available without specifying a compatible coherent receiver and DSP architecture. The recommended baseline is controlled launch power, dispersion-aware link design and receiver equalization appropriate to the selected detection format. DBP may be evaluated as an optional enhancement if the receiver architecture supports it.

5.5 Standards Verification

ITU-T G.957 specifies optical interface characteristics for relevant synchronous digital hierarchy systems. . Verification should include the relevant wavelength range, transmitter power, spectral characteristics, modulation requirements, receiver compatibility and interface limits. IEEE 802.3bs and IEEE 802.3bm define Ethernet physical-layer implementations, including high-speed optical interfaces. These references must be checked against the actual system architecture and applicable clauses. They do not automatically establish compliance for an arbitrary eight-channel DWDM transmitter.

ANSI Z136 requirements must be addressed separately for laser safety.

5.6 Photonic Integration

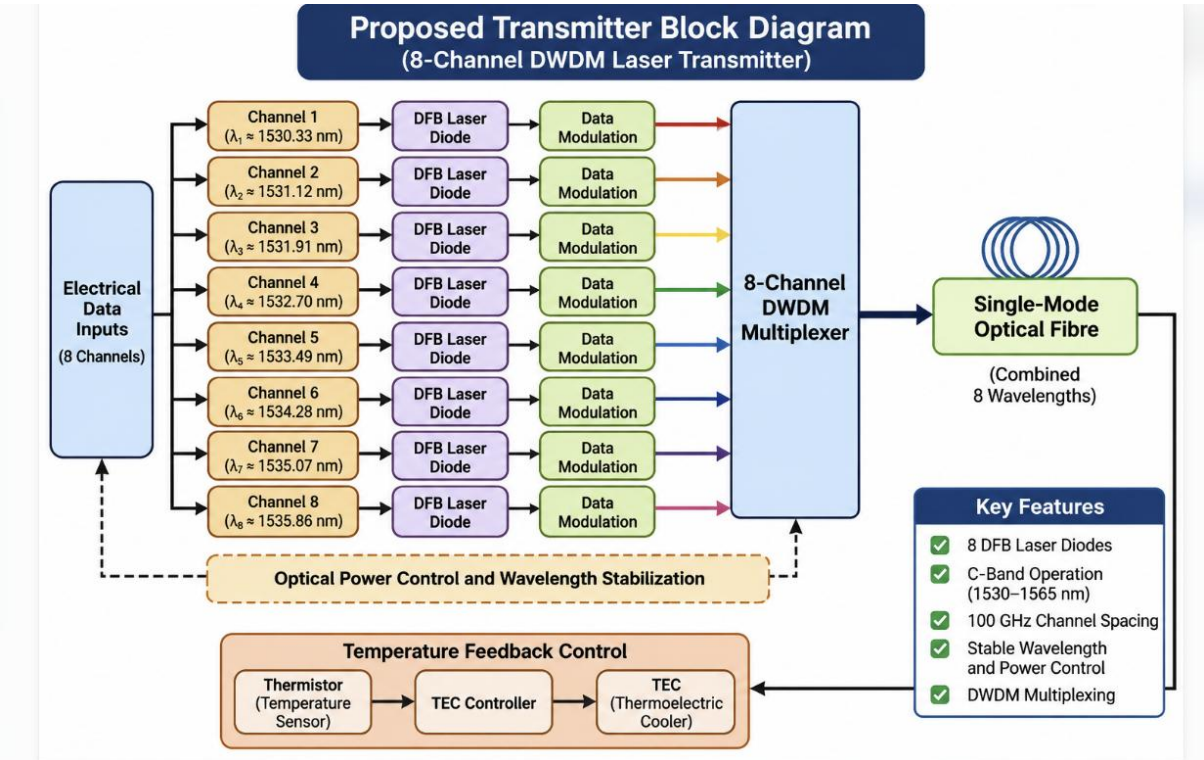
Implementation	Advantages	Limitations
Discrete components	Flexible selection, accessible testing and replacement	Larger footprint, more interconnections and coupling interfaces
Monolithic InP integration	Compact optical integration and reduced interconnect count	Process complexity, thermal design and fabrication cost
Si-photonics with hybrid laser	Potential scalability, compact routing and integration with electronic control	Requires hybrid laser integration and careful coupling and thermal management

For an eight-channel transmitter, silicon photonics with hybrid-integrated DFB lasers is recommended as the development target because it can support compact multi-channel routing and scalable integration. This recommendation is conditional on fabrication access, thermal performance, channel isolation, reliability and demonstrated optical performance. A discrete prototype is appropriate for initial laboratory validation.

Design Decision 4: I recommend a silicon-photonics transmitter with hybrid-integrated, temperature-controlled DFB lasers, external modulators and appropriate receiver-side compensation. This approach offers a scalable implementation while preserving the spectral and modulation control needed for DWDM operation. Full multi-channel DBP should remain optional until its receiver compatibility and processing requirements are established.

6. FINAL TRANSMITTER DESIGN AND VERIFICATION

6.1 Proposed Transmitter Block Diagram



The diagram represents the proposed architecture. The exact amplifier arrangement, multiplexer insertion loss, receiver design and compensation implementation must be established during detailed system design.

6.2 Final Parameter Table

Design parameter	**Proposed value or decision**
Optical source	DFB laser per channel
Calculated emission wavelength	1550 nm
Number of channels	8
Channel spacing	100 GHz
Per-channel data rate	25 Gb/s NRZ
Aggregate data rate	200 Gb/s
Calculated laser output	8.4 mW or 9.24 dBm

Specified fibre launch power	2 mW or 3.01 dBm per channel
Modulation bandwidth	20 GHz
Required approximate bandwidth	17.5 GHz
Bandwidth margin	2.5 GHz
Wavelength control	TEC with feedback
Temperature tolerance from drift model	Approximately ±1 °C
Modulation configuration	External Mach–Zehnder modulation
Effective fibre length	21.72 km per span
SPM phase shift	0.0565 rad per span
Six-span SPM estimate	0.339 rad
SBS threshold estimate	1.55 mW
SRS threshold estimate	983 mW
Dispersion length	11.25 km
Fundamental-soliton peak power	68.38 mW
Soliton order at 2 mW	0.171
Soliton transmission	Not recommended at the stated power
Integration target	Silicon photonics with hybrid lasers
Standards and safety	Verify applicable interface clauses and ANSI Z136 requirements

7.CONCLUSION AND FINAL ENGINEERING RECOMMENDATION

Temperature stabilization is essential because the predicted uncontrolled wavelength drift is 4 nm, which exceeds the allowed ±0.1 nm tolerance. External modulation and TEC feedback are therefore recommended to reduce chirp-related penalties and stabilize the wavelength grid. Consequently, soliton transmission is not recommended under the given conditions.

The analysis supports the selection of temperature-controlled DFB lasers operating around 1550 nm for the eight-channel DWDM transmitter. The calculated emission wavelength is 1550 nm, and the internal quantum efficiency is 80%. At the specified bias current, the simplified output-power model gives 8.4 mW, while the required NRZ bandwidth is approximately 17.5 GHz. The candidate 20 GHz bandwidth provides a 2.5 GHz margin. Dispersion-aware engineering and receiver-side compensation are more appropriate for the proposed baseline system

The soliton analysis produces a dispersion length of 11.25 km and a soliton order of approximately 0.171 at the specified launch power.